

Student Exam Score Prediction Using Machine Learning

Introduction :

Machine Learning is one of the most important technologies in today's world. It helps computers learn patterns from data and make predictions without being explicitly programmed. Machine Learning is widely used in education, healthcare, banking, business, and many other fields.

In the education sector, predicting student performance is very useful for teachers and institutions. By analyzing student study habits and attendance, we can estimate the student's exam performance. This helps teachers identify weak students and provide guidance at the right time.

The main aim of this project is to predict the exam score of students using Machine Learning algorithms. In this project, factors such as study hours, sleep hours, and attendance are used as input features to predict the final exam score.

This project uses different Machine Learning algorithms like Linear Regression, Random Forest Regression, and XGBoost Regression to compare performance and obtain accurate predictions. The models are evaluated using the R2 Score method. Finally, the trained model is saved using a pickle (.pkl) file and deployed using Streamlit, which allows users to enter values and predict student scores through a simple interface.

● Objective

The main objective of this project is to develop a Machine Learning model that can predict student exam scores based on different student-related factors.

The specific objectives of this project are:

To collect and analyze student-related data.

To preprocess the dataset for training the Machine Learning model.

To implement different regression algorithms.

To compare the performance of various Machine Learning models.

To evaluate model accuracy using evaluation metrics.

To save the trained model using Pickle.

To deploy the project using Streamlit.

To create a user-friendly prediction system.

This project helps in understanding how Machine Learning can be applied in the education field to improve student performance analysis.

Dataset

The dataset used in this project contains student information related to study habits and attendance.

The dataset includes the following columns:

Column Name

Description

study_hours

Number of hours studied by the student

sleep_hours

Number of sleeping hours

attendance

Student attendance percentage

score

Final exam score

The dataset is stored in CSV format and used for training and testing the Machine Learning models.

● Algorithms Used :

1. Linear Regression :

Linear Regression is a supervised Machine Learning algorithm used for predicting continuous values. It establishes a linear relationship between input variables and output variables.

In this project, Linear Regression predicts student scores based on study hours, sleep hours, and attendance.

Advantages:

Simple and easy to implement

Fast training process

Good for small datasets

2. Random Forest Regression :

Random Forest is an ensemble learning algorithm that uses multiple decision trees to improve prediction accuracy.

It combines the predictions of several trees and provides more stable and accurate results.

Advantages:

Higher accuracy

Reduces overfitting

Handles complex data efficiently

3. XGBoost Regression :

XGBoost stands for Extreme Gradient Boosting. It is an advanced Machine Learning algorithm known for high performance and speed.

XGBoost improves prediction results by correcting previous model errors during training.

Advantages:

High prediction accuracy

Efficient and fast

Better optimization techniques

● Model Evaluation

The Machine Learning models are evaluated using the R2 Score metric.

R2 Score :

R2 Score measures how well the model predicts the output values.

R2 Score close to 1 indicates better performance.

Higher R2 Score means better prediction accuracy.

The performance of Linear Regression, Random Forest, and XGBoost models is compared to identify the best model.

Random Forest achieved better performance compared to other models in this project.

Overfitting Check :

Overfitting occurs when the model performs well on training data but poorly on testing data.

To avoid overfitting:

The dataset was split into training and testing sets.

Different Machine Learning algorithms were compared.

Model evaluation metrics were used.

This helped in selecting the best-performing model with balanced accuracy.

● Deployment

The final trained model is deployed using Streamlit.

Streamlit is a Python framework used to create web applications for Machine Learning projects.

The deployed application allows users to:

Enter study hours

Enter sleep hours

Enter attendance percentage

Predict student exam score instantly

The trained model is stored using a .pkl file and loaded into the Streamlit application.

● Technologies Used

Python

Pandas

Scikit-learn

XGBoost

Streamlit

Pickle

Google Colab

Conclusion :

This project successfully predicts student exam scores using Machine Learning techniques. Different regression algorithms such as Linear Regression, Random Forest, and XGBoost were implemented and evaluated.